

Improved strength of alkali activated materials based on construction and demolition waste with addition of rice husk ash

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ABSTRACT

A promising solution to the environmental problem of managing a huge amount of construction and demolition waste (CDW) is its use as precursor in the production of new alkaline-activated cements. Specifically, the main objective of this investigation is to increase the strength of alkaline activated cements synthesized from un-separated CDW of low reactivity due to a high presence of crystalline phases in their composition. The effect of two factors is analyzed: the partial substitution of CDW by rice husk ash (RHA) (20–40 wt%) as well as the modulus of the alkaline activator used ($M_s = \text{SiO}_2/\text{Na}_2\text{O}$ of 0.5, 1.0 and 2.0). Physical, mechanical and thermal testing, Fourier transform infrared spectroscopy (FTIR), X-ray diffraction (XRD) and scanning electron microscopy with energy dispersive X-ray spectroscopy (SEM-EDS) are used to characterize alkali-activated cements. Results show that the incorporation of increasing amounts of RHA and the increase in the modulus of the activator produce changes in the general structure of the gel, obtaining a greater amount of N-A-S-H geopolymeric gel with respect to the main reaction product, the C-A-S-H gel obtained in the control cements containing CDW as the only precursor. The new reaction products and the change in the microstructure make it possible to achieve a compressive strength of 59.6 MPa at 28 curing days in the materials with 40 wt% RHA and $M_s = 2.0$ compared to 18.5 MPa for the material developed exclusively from CDW. Therefore, this study validates the use of CDW together with RHA as precursors in the production of alkaline activated materials with improved technological properties. These green cements would help to reduce the presence of CDW and RHA in landfills and would generate new environmentally sustainable materials.

1. Introduction

Construction and demolition waste (CDW) is one of the most abundant types of waste generated worldwide. In Europe, it accounts for approximately one-third of the generated waste and constitutes one of the priority material streams when establishing circular economy strategies [1]. CDW is a valuable material that is not always identified, reused and recycled, with the challenge of exhibiting a variable average composition. It encompasses all waste produced during the construction and demolition of buildings and infrastructure and thus it contains (excluding soils) a wide variety of materials such as concrete, bricks,

wood, glass, metals, plastics, asphalts and solvents. Since backfilling and landfilling operations are still significant for this waste [1,2] inadequate management and disposal of these materials can lead to irreversible damage to the environment and ecosystems, consequently compromising the health and well-being of individuals [3,4].

In light of the importance of the reduction, reuse and recycling of CDW, significant progress has been made in recent years in research focused on improvements in its management and sorting, as well as in the implementation of models to predict and identify material flows generated and their future demand, taking into account the local characteristics of the construction process. Thus, a proper management and

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knowledge of the CDW produced in every construction/demolition process enable the generation of material flows with a more homogeneous composition over time. This increases the possibilities for reuse and recycling in various processes within the construction sector itself [5–14].

These improvements in the management of inert materials currently sorted from CDW produce, as the main recycling product, a mineral or stony fraction of coarse-sized particles (aggregates) whose composition and primary use depend on the regional characteristics of the treated waste. For instance, in southern Europe where many building elements are ceramic-based, a fraction of recycled aggregates (primarily composed of concrete) is distinguished from mixed recycled aggregates (with significant percentages of masonry rubble), with this latter category accounting for 70% of total recycled aggregates in Spain according to [15]. Among the main uses of these recycled aggregates is the replacement of natural aggregates for concrete production and for road base/sub-base filling [7,16–19]. Generally, the substitution of natural aggregates with recycled aggregates results in mechanical deterioration in the products and applications in which they are incorporated, potentially limiting their utilization despite the environmental benefits they offer. Thus, the most common usage in Spain of mixed recycled aggregates with an important ceramic fraction is predominantly in applications with low added value, such as in unpaved rural roads with low daily heavy traffic [15].

A promising alternative for the recovery of CDW is its utilization in the production of new sustainable construction products. In particular, the partial replacement of Portland cement with inert fillers and inorganic wastes [20,21] or the development of alkali-activated binders or geopolymers are described as sustainable alternatives for the building sector. Concerning this last alternative, under the designation of alkali-activated binders or geopolymers, a multitude of new materials have been proposed, with the main objective of substituting Portland cement in different applications for construction [22–25]. These alkali-activated materials have in common their production through the chemical reaction between alkalis and aluminosilicates, which may come from natural sources (clays) or from waste sources (ashes, slags, glasses, etc.). The goal is the formation of reaction products or gels characterized by a partially or fully amorphous polymeric structure that consists of Si-O-Al links with cementing properties [26]. Thus, the portion of construction and demolition waste (CDW) based on mortar and ceramics, composed of silicon materials more or less enriched in aluminum and calcium, is considered ideal for the formulation of this family of materials. It has been investigated both as aggregate for mortars and concretes based on new alkali-activated cements and in the manufacturing of the binder itself. This CDW recovery alternative presents as technological and environmental advantage that alkali activated materials are synthesized at relatively low temperatures (below 100 °C), which allows for a significant reduction in both greenhouse gas emissions and energy consumption in relation to ordinary Portland cement (OPC) manufacture. Additionally, it reduces the consumption of natural raw materials as it reuses solid waste to obtain materials that can provide performance comparable to OPC [27,28].

The most evident valorization alternative of CDW in alkali-activated materials is as a partial or total substitute for natural aggregates in geopolymeric concretes and mortars and stabilization of base/subbase road pavement layers. In general, different authors have described experiences with these recycled aggregates using the particle size distribution of sand (<2 mm) [29–31] or gravel (up to 20 mm) [32–34] and employing different binder mixtures that mainly contain precursors such as metakaolin, slag and fly ash. However, many of these works do not demonstrate the durability of the obtained products, which, due to the development of more porous materials, can be reduced [27,33,35]. On the contrary, in these lighter materials, a positive effect of slag usage is generally described, and the ceramic fraction in the mixed recycled aggregates shows potential for activation in the alkaline medium and develops better interfaces between the aggregate and the matrix [29,

36]. Thus, comprehensive utilization of CDW both as aggregate and in the binder itself has been investigated in different systems, which usually involve using a formulation based on different CDW components (concrete, ceramics, and glass), this demanding prior separation. Optimization of these last materials entails the use of slag as supplementary cementitious material and the addition of $\text{Ca}(\text{OH})_2$ as a complement to the usual alkaline activator consisting of mixtures of $\text{NaOH} + \text{Na}_2\text{SiO}_3$ [37,38].

In relation to the use of CDW solely as a binder precursor, various strategies have been described to achieve sufficient strength in the developed materials. Some authors propose the separation of the waste into its different fractions (brick, ceramics, concrete, glass, etc.) [39–43], while Mahmoodi et al. [44], and Tan et al. [45], suggest a possible recombination of these fractions following a controlled formulation. These works generally observe that the brick fraction produces higher mechanical strength, and that proper formulation allows for the adjustment of the setting time by controlling the Si/Al, Na/Si, and Ca/Si ratios. In these studies, the predominant formation of C (A)SH gel is observed and the formation of NASH gel is primarily associated with the remnants of red bricks. However, considering the technological difficulty of sorting and separating these components, other studies focus on the valorization of unseparated CDW, investigating its mixture with other precursors (mainly fly ash, ground granulated blast furnace slag and metakaolin) to increase the strength of the developed materials [46,47]. Another type of precursor mixtures includes the addition of ordinary Portland cement to obtain hybrid cements [41].

In this last context of formulating blends of unselected CDW with other residues to enhance the properties of alkaline activation binders, the present study proposes the addition of rice husk ash (RHA). Rice husk is separated from the rice grains during the milling process and, therefore, represents a significant, abundant, and readily available biomass source, considering that global rice production reached approximately 760 million tons in 2017 [48]. Rice husk holds great potential for bioenergy generation due to its high calorific value (one ton of rice husk can generate 800 kW), and approximately 20–25% of its weight is converted into ash upon incineration [49]. Despite their potential applications in construction products [50] only a small fraction of these ashes is valorized.

RHA is primarily composed of silica, in either amorphous or crystalline form depending on combustion conditions [50,51], making it of interest for alkaline activation binders, both as a precursor component of the binder and as a potential alkaline activator. Consequently, there is currently a promising line of research for the sustainable production of alkaline activators, such as sodium silicate, from the dissolution of RHA in NaOH [52]. According to various studies conducted with different precursor materials (fly ash, blast furnace slag, metakaolin), the total or partial replacement of commercial sodium silicate with this alternative solution enhances the material's strength and significantly reduces its environmental and health impact [53–55]. In general, higher levels of soluble silica are associated with improved mechanical properties of the mixtures [56].

On the other hand, for the use of RHA as a precursor for alkaline activation binders, even though this residue is an excellent source of amorphous silica, its alumina content is extremely low. Therefore, various studies have evaluated the potential of mixtures with other raw materials such as aluminum anodizing sludge [57] or fly ash [58,59]. In the latter, it has been described that the increase in RHA content increases the compressive strength of the material and results in a microstructure with the presence of geopolymerization reaction products and a small amount of unreacted RHA and fly ash particles. In this regard, Nuaklong et al. [60] investigated the development of geopolymeric concrete based on the incorporation of RHA on fly ash binders and the addition of recycled aggregates from CDW.

2. Significance of the research

The main innovation of this research is the combination of waste proposed as precursor for alkaline activation binders. Thus, to the best of the authors' knowledge, CDW and RHA mixtures have not been previously evaluated for this application. Only the use of recycled CDW aggregates has been proposed for the reinforcement of concrete based on other types of precursor mixtures with RHA. In the combination evaluated in this study, RHA is used as a source of additional amorphous silica to complement the aluminosilicates in the precursor CDW, which allows enhancing the process of alkali activation.

The main goal of the paper is to establish how RHA additions effectively improve strength of final binders. To do this, from previous studies [61], a geopolymeric material based exclusively on CDW from southern Spain as precursor, with a highly crystalline structure, is taken as a starting point. The high crystallinity makes this residue relatively inert to alkaline attack, making it a candidate of interest for formulate admixtures with other precursors such as RHA to increase strength. The expected synergistic effect of the joint recovery of both wastes can contribute to reducing the environmental impact of their disposal in landfills and to obtaining new binders with a lower environmental footprint.

3. Materials and Methods

3.1. Raw materials specifications

The main raw material utilized as aluminosilicate precursor in the manufacturing of alkali-activated binders was construction and demolition waste (CDW). The Andalusian Association of Construction and Demolition Waste Management Companies (AGRECA) provided the CDW. Specifically, it was selected a mixed aggregate material with concrete and ceramic components originated from a large urban area that exhibited significant uniformity in its properties. Additionally, Herba Ricemills S.L., a rice production industry located in Seville, Spain, supplied the required rice husk ash (RHA).

Experimental procedure to characterize and condition raw materials is summarized in Fig. 1. Starting CDW precursor has been described elsewhere [61] showing a high crystallinity. It is a recycled mixed graded aggregate for sub-base with a particle size range 0–40 mm and a composition of mainly concrete debris with ceramic remains. It was necessary to subject them to crushing and milling. This process aimed to produce a powder with particle size suitable for passing through a 150 μm sieve. Starting RHA also has been previously investigated for the production of ceramic bricks [51]. This waste exhibited heterogeneity in particle size. Hence, in order to achieve uniform particle size, it was subjected to ball milling up to a maximum size of 150 μm . In Fig. 2, the particle size distribution of both precursors after ball milling is shown. Morphology of the raw particles was observed using a Carl Zeiss

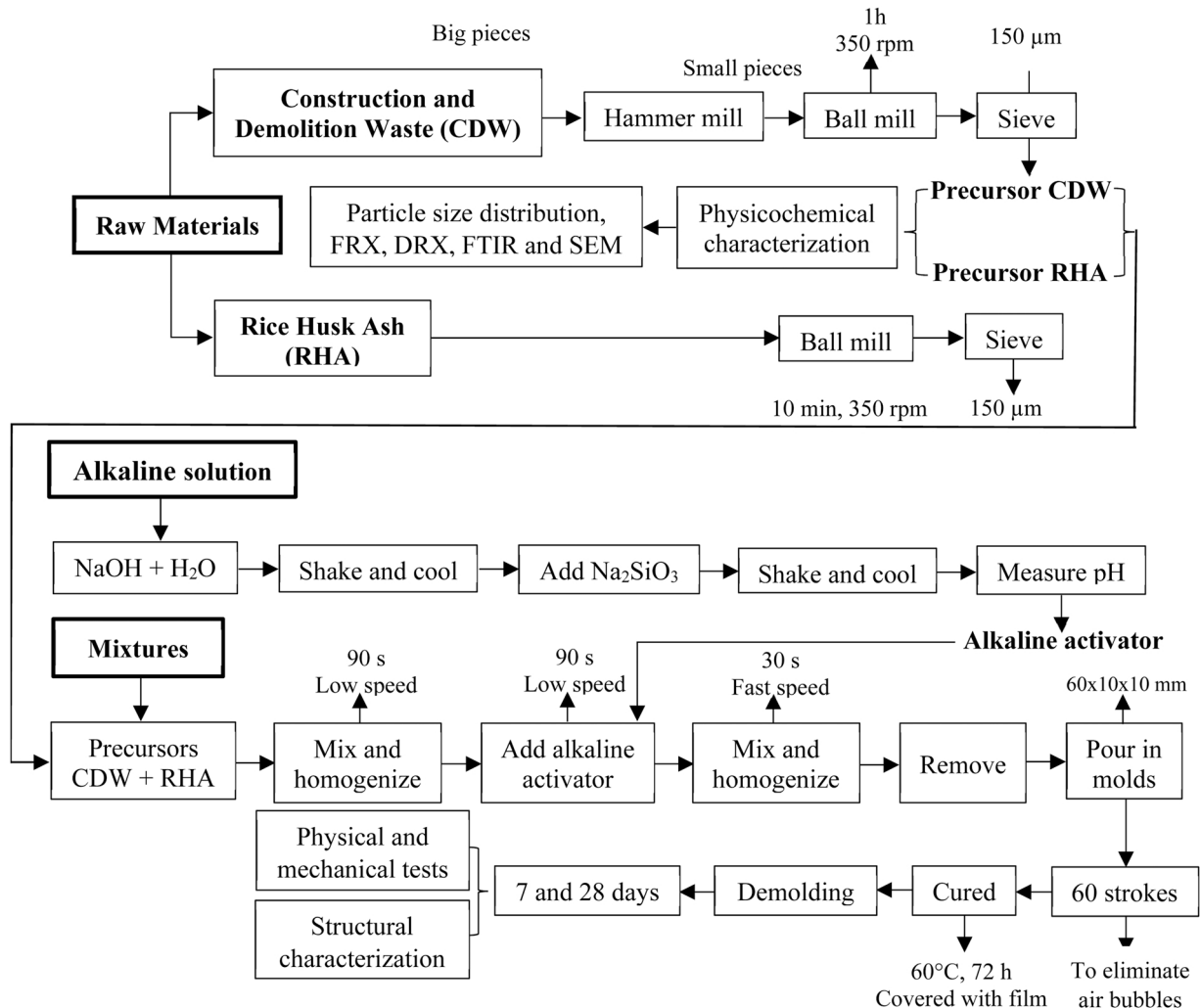


Fig. 1. Diagram of the manufacturing process and characterization carried out.

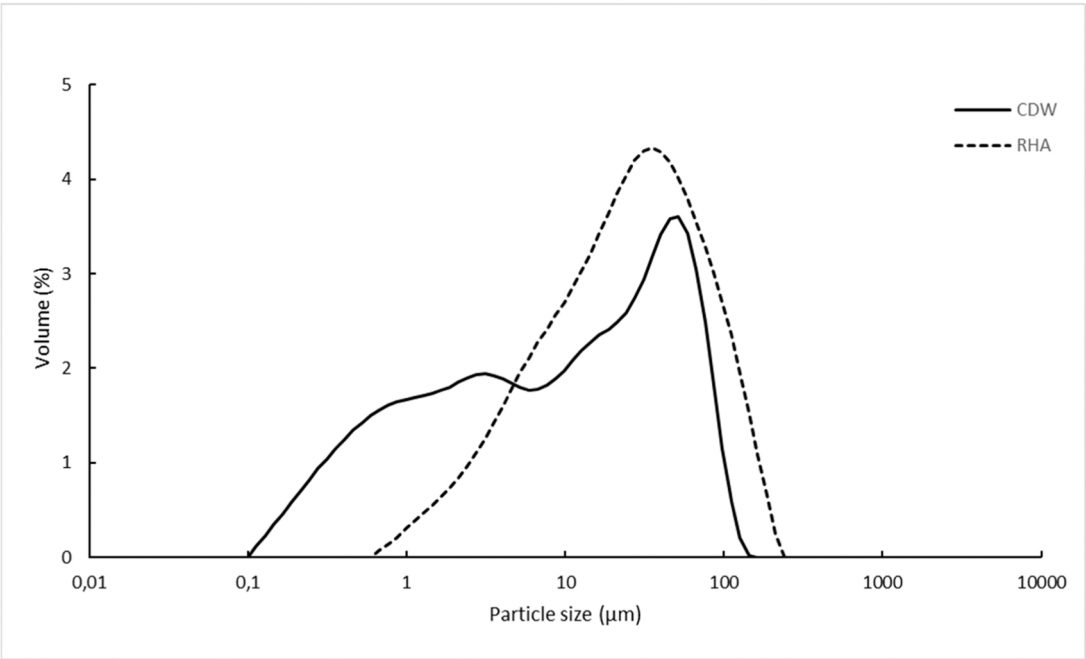


Fig. 2. Particle size distribution of both precursors (CDW and RHA) after ball milling.

Scanning Electron Microscopy (SEM), model Merlin, assisted by Energy Dispersive X-ray Spectroscopy (EDS) Oxford Inca Energy 350X-MAX 50. The Fig. 3 displays characteristic SEM images of particles from both milled raw materials (CDW and RHA). Microanalysis on these images confirms the silicon-rich composition of RHA, and Si, Al, Ca, and Fe as

the primary elements in the case of CDW. The chemical composition of the raw materials was analyzed by X-ray fluorescence (XRF) technique employing a Philips Magix Pro model PW-2440 equipment. Table 1 shows that CDW primarily consists of SiO₂ (51.7 wt%) and CaO (18.9 wt%), with a relatively low content of Al₂O₃

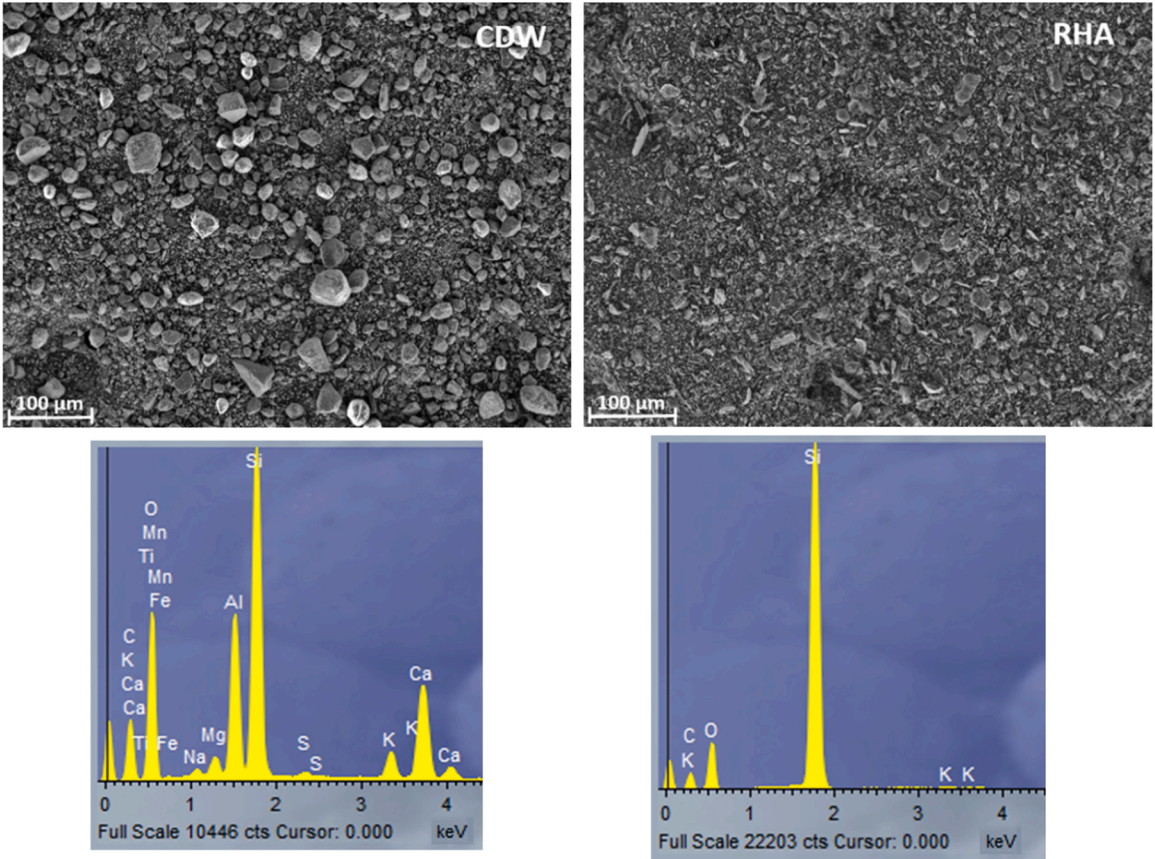


Fig. 3. SEM images and EDS analysis of precursors CDW and RHA.

Table 1

Chemical composition (wt%) of construction and demolition waste (CDW) and rice husk ash (RHA).

Material	wt%										
	SiO ₂	CaO	Al ₂ O ₃	Fe ₂ O ₃	MgO	K ₂ O	SO ₃	Na ₂ O	TiO ₂	P ₂ O ₅	LOI
CDW	51.7	18.9	5.9	2.5	2.1	1.4	0.6	0.6	0.3	0.1	15.5
RHA	73.6	0.8	-	0.3	0.7	1.6	0.1	0.1	-	1.8	20.8

(5.9 wt%). Other oxides present in approximately 2 wt% quantities include Fe₂O₃, MgO, and K₂O. On the other hand, the main components of RHA are SiO₂ (73.6 wt%), with a minimal CaO content (0.8 wt%) and no detectable presence of Al₂O₃, with percentages lower than 2 wt% for other components like K₂O (1.6 wt%) and P₂O₅ (1.8 wt%). RHA residue exhibit higher loss by ignition (LOI) values (20.8 wt%) than CDW (15.5 wt%). These values are attributed primarily to the presence of organic matter and carbonates in RHA and CDW respectively.

The X-Ray Diffraction technique (XRD) was employed to determine the crystalline mineralogical phases present in the raw materials. An Empyrean X-ray powder diffraction equipment with a PIXcel-3D detector from PANalytical was used for this analysis. The identification of crystalline phases was conducted using HighScore software. Fig. 4 illustrates that the XRD pattern of the CDW exhibits a semicrystalline nature, with the dominant crystalline phase identified as quartz (Ref. code: 96-500-0036). Additionally, other phases were identified, including sodium-calcium feldspars belonging to the plagioclase group (albite and anorthite) (Ref. code: 96-155-7000 and 96-100-0035), phyllosilicates such as illite (Ref. code: 96-901-3719), as well as calcite (Ref. code: 96-901-6707) and dolomite (Ref. code: 96-900-4931). On the other hand, the XRD pattern of the RHA residue displays a broad halo in the range of 15–30° 2θ, indicating the presence of an amorphous phase in the material. Additionally, crystalline peaks corresponding to cristobalite (Ref. code: 96-900-9687) were observed.

3.2. Alkali-activated synthesis

To initiate the processing, alkaline activating solutions were prepared 24 h prior to mixing with the precursors (Fig. 1). Three different solutions were prepared, containing 25 wt%, 50 wt%, and 75 wt% of sodium silicate in the mixture. The remaining percentage consisted of an

8 M NaOH solution, ensuring an activator modulus, $M_s = \text{SiO}_2/\text{Na}_2\text{O}$ of 0.5, 1.0, and 2.0, respectively (Table 2). To prepare the 8 M NaOH solution, sodium hydroxide pellets supplied by Panreac S.A. (Spain) with a purity of 98% were dissolved by magnetic stirring in distilled water. Subsequently, when the NaOH solution had cooled, the corresponding quantity of a commercial sodium silicate solution, provided by Panreac S.A. (Spain), with a composition of 29.2 wt% SiO₂, 8.9 wt% Na₂O, and 61.9 wt% H₂O, was added slowly and continuously. The activating solutions were then allowed to cool at room temperature, resulting in a pH range of 12–13.

In the study, different proportions of RHA (0 wt%, 20 wt%, and 40 wt%) replaced CDW so starting solid compositions were named as 100CDW, 80CDW, 60CDW, and the pastes were prepared by adding the activating solution to the solid raw materials (Fig. 1). As shown in Table 2, nine mixtures were proposed, varying the percentage of sodium silicate (25%, 50%, and 75%) in the activator solution for each solid composition (100CDW, 80CDW, and 60CDW). The amount of water added to each mixture was adjusted to achieve similar workability in all pastes. Therefore, Table 2 presents the water/binder ratio (w/b), which indicates the total mass of water in the mixture (including the water in the activator solution) divided by the mass of the solids (CDW+RHA).

The mixing process was conducted using a Proeti planetary mixer, following the same sequence for all compositions. Initially, only the solid raw materials were mixed and homogenized for 90 s at a low speed. Afterward, the alkaline activator was added, and the paste was mixed for additional 90 s at a slow speed. Finally, any material adhering to the vessel walls was removed, and the mixture was further mixed for 30 s at a high speed. Following this kneading stage, the fresh mixtures were poured into rectangular stainless steel molds measuring 60×10×10 mm and circular ones with a diameter of 55 mm and a thickness of 15 mm treated with oil. The filled molds were subjected to

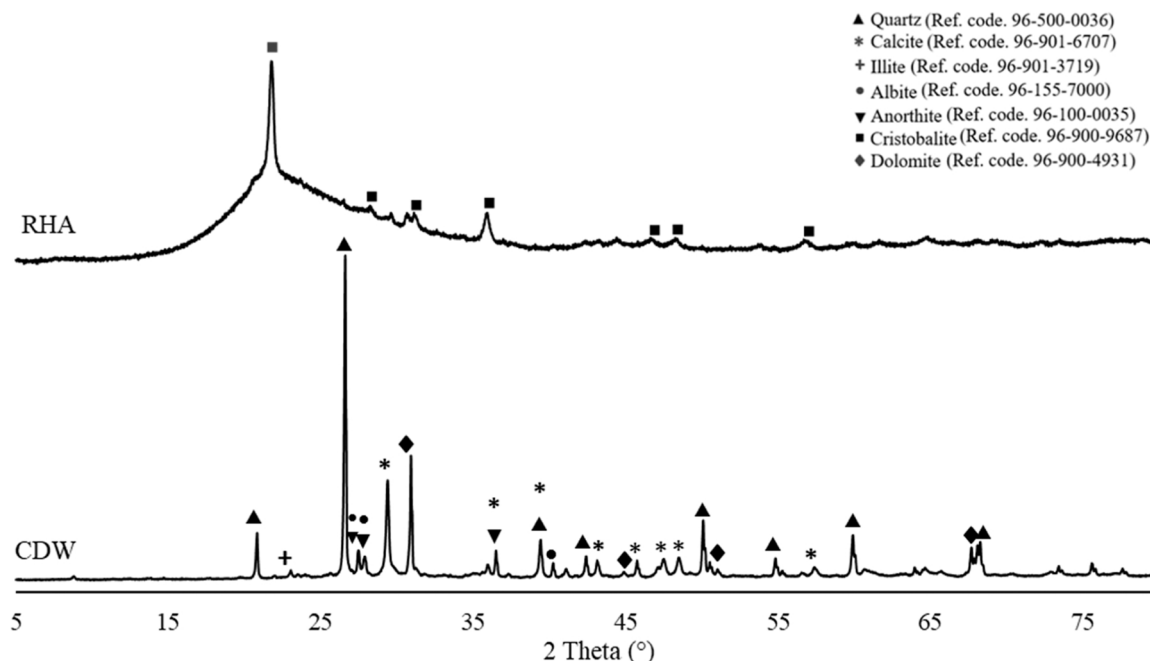
**Fig. 4.** XRD of precursors: CDW and RHA.

Table 2

Component proportions in the manufactured mixtures.

Mix	CDW (g)	RHA (g)	H ₂ O (g)	NaOH (g)	Na ₂ SiO ₃ (g)	Na ₂ SiO ₃ (wt%)*	Ratio w/b	Ms (SiO ₂ /Na ₂ O)
100CDW	400	0	90.5	29.5	40	25	0.4	0.5
	400	0	60.3	19.7	80	50	0.4	1.0
	400	0	30.2	9.9	120	75	0.4	2.0
80CDW	320	80	124.4	40.6	55	25	0.55	0.5
	320	80	82.9	27.1	110	50	0.55	1.0
	320	80	41.5	13.5	165	75	0.55	2.0
60CDW	240	160	235.9	77.2	104.3	25	1.043	0.5
	240	160	157.3	51.4	208.6	50	1.043	1.0
	240	160	78.6	25.7	312.9	75	1.043	2.0

w: water; b: binder

*wt% in the alkaline activator solution

60 strokes on a Proeti shaking table to eliminate air bubbles and enhance compaction. The specimens were then cured at 60 °C for 72 h while wrapped in an adhesive film. This temperature is in the range of conditions reviewed for curing geopolymers [62]. After the 72 h period, the samples were removed from the molds and kept at room temperature until the characterization tests were conducted at 7 and 28 days. For each mixture, 24 rectangular specimens were fabricated. These samples were labeled as x-CDW-y, where x denotes the CDW content (100, 80, or 60 wt%, with the remainder being RHA), and y indicates the activator modulus, Ms (0.5, 1.0, 2.0).

3.3. Characterization of alkali-activated materials

The bulk density of the samples at 28 days was determined using the Archimedes method following the UNE-EN 1015 10:2000/A1:2007 standard [63]. The average of five determinations was taken for each sample and error is the standard deviation. The manufactured bars were dried until a constant weight was achieved. Subsequently, the specimens were immersed in water at room temperature for 24 h in a container. These submerged bars were weighed using a hydrostatic balance, and their mass was recorded. They were then gently dried with a damp cloth to remove surface water, and the saturated weight of the test pieces was noted. The real density was measured using a pycnometer with fine crushed specimens that had a particle size smaller than 63 µm. Ethanol was used as the displacing liquid.

Water absorption (wt%) was determined by immersing dry specimens in water at room temperature for 24 h and comparing the wet mass with the dry mass. Apparent open porosity (in vol%) was determined by calculating the weight difference between the saturated mass and the dry mass relative to the external volume, following the calculations from ASTM standard C373 [64]. The total porosity is calculated as the percentage ratio between the total volume of the pores (open and closed) and the apparent volume of the specimen, using Eq. (1) and closed porosity was calculated from the difference between total and open porosity [65].

$$\text{Total porosity} = \left(1 - \frac{\text{Bulk density}}{\text{Real Density}}\right) \cdot 100 \quad (1)$$

For the flexural strength, five bar specimens of each composition were tested at 7 and 28 days of curing. The reported results are the average and the standard deviation. The tests were conducted using an MTS Insight 5 machine (Eden Prairie, MN, USA) with a capacity of 5 kN and a displacement speed of 1.0 mm/min, following the UNE-EN 1015-11:2020 standard [66]. The compressive strength was determined according to the UNE-EN 1015-11:2020 standard [60], using an MTS 810 machine. The halves from the flexural test were subjected to progressively increasing normal stress (machine displacement speed of 2.0 mm/min) until they fractured. Reported values are the average of five determinations at 7 and 28 days and errors are the standard deviations.

The thermal conductivity was determined using a TA Instruments-

Waters LLC® heat flow meter, model FOX 50, following the ISO-8302:1991 standard [67]. This test was performed on the circular specimen with a diameter of 55 mm and a thickness of 15 mm at 28 days of curing.

Regarding compositional characterization of the obtained reaction products, XRD was applied to ground and milled specimens ($\leq 100 \mu\text{m}$) and SEM-EDX to fracture surfaces of specimens after compressive strength, using the same equipment and operating conditions as for the raw materials. The identification of functional groups was carried out by attenuated total reflectance Fourier transform infrared spectroscopy (ATR-FTIR) using a Vertex 70 Bruker equipment in the range 4000–400 cm^{-1} .

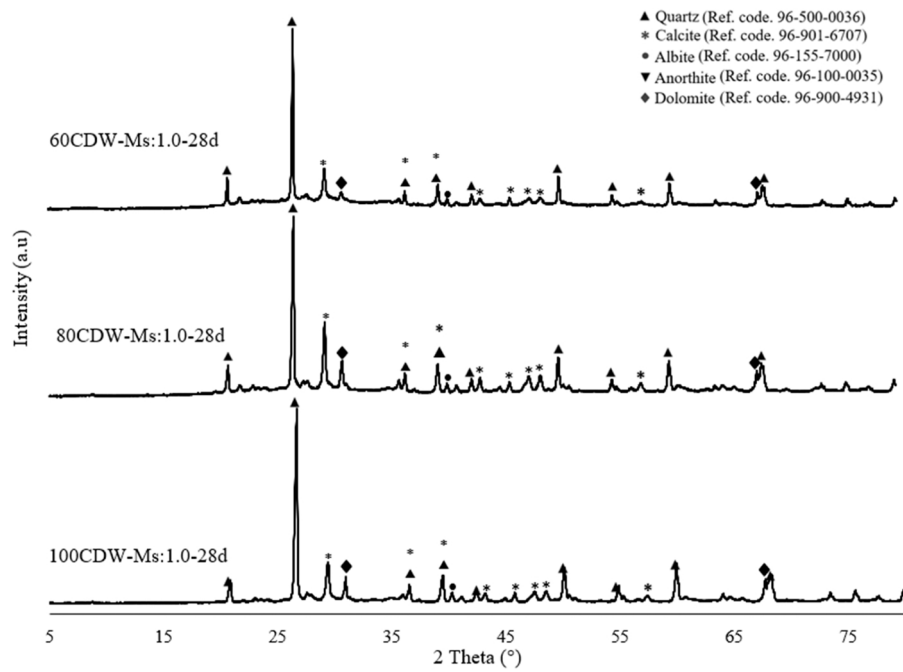
4. Results and Discussion

4.1. Forming, composition and microstructure

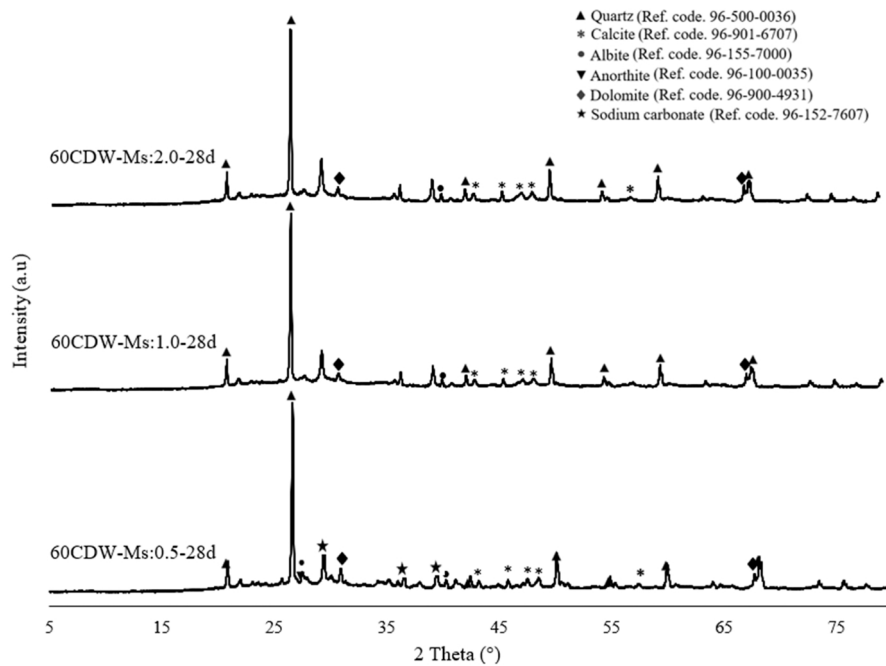
Concerning the characteristics of the solid raw materials, as outlined in a previous publication [61], the initial Construction and Demolition Waste (CDW) used in this study constitutes a mixed aggregate material containing both concrete and ceramic fractions. This particular unselected composition is less widely accepted for sub-base applications compared to its concrete-rich counterpart. This previous investigation demonstrated that this type of CDW exhibits diminished strength when subjected to alkaline activation due to its pronounced crystalline nature (Fig. 4). In relation to Rice Husk Ash (RHA), which is primarily comprised of amorphous silica, it was utilized in its natural form following suitable milling procedures. No calcination or heat treatment was applied to eliminate residual unburned material (as indicated by the LOI in Table 1), aiming to minimize the process's energy footprint for alkaline activation, and as reported by Shearer et al. [68], in the case of biomass-coal fly ash. The substantial organic content within the ash is attributed to uncontrolled burning temperatures during rice husk incineration [51], which imparts the distinctive black hue to this waste material.

Starting from conditioned raw materials, pastes preparation and molding of the specimens was successfully carried out for all the materials proposed in this research, owing to the favorable workability of the pastes. Nonetheless, pastes incorporating RHA display decreased workability attributed to the presence of carbonaceous content in the ashes (as indicated by LOI in Table 1), necessitating a higher water-to-binder ratio compared to pastes formulated exclusively with CDW (w/b=0.4, Table 2) in order to empirically achieve equivalent workability.

To investigate the impact of incorporating RHA in the alkaline activation process of CDW, the initial characterization phase involved analyzing the composition and microstructure of the synthesized materials. Fig. 5a illustrates the XRD pattern of alkali-activated binders after 28 days of curing for an Ms value of 1.0. It can be observed that these materials exhibit similar diffraction peaks corresponding to the crystalline phases present in the CDW precursor (Fig. 5a). This observation highlights the low reactivity of the starting precursor [61]. Across



(a)



(b)

Fig. 5. a. XRD of alkali activated cements at 28 days of curing with $M_s = 1.0$. b. XRD of 60CDW-40RHA at 28 days of curing with $M_s = 0.5, 1.0$ and 2.0 .

all alkali-activated binders, strong diffraction peaks corresponding to quartz, calcite, and dolomite, present in the CDW precursor, indicate that these phases are inert and participate to a lesser extent in the alkali activation process. However, the intensity of the diffraction peaks decreases as increasing amounts of RHA are incorporated, suggesting that certain crystalline compounds, primarily sodium-calcium feldspars and

phyllosilicates, partially dissolve and take part in the alkali activation reaction. Furthermore, a comparison of the XRD patterns of the alkali-activated materials with those of the CDW precursor reveals the emergence of a broad asymmetric "hump" in the 2θ range of $20\text{--}35^\circ$, more pronounced in the binders containing RHA. This suggests the formation of new amorphous cementitious products during the alkali

activation reaction, possibly related to the generation of N-A-S-H (Sodium Alumino-Silicate Hydrate) and C-(A)-S-H (Calcium-Aluminosilicate Hydrate) geopolymers [69,70]. Additionally, an incipient semi-crystalline phase appears near $30^\circ 2\theta$, that can be attributed to the formation of low-calcium C-S-H gels [71,72]. On the other hand, in Fig. 5b, the influence of the activator modulus, M_s , is depicted for the binder 60CDW-40RHA as a representative case. It can be observed that alkali-activated cements manufactured across the entire M_s range show no significant changes, except for $M_s=0.5$, where more intense diffraction peaks of crystalline phases are observed, indicating reduced effectiveness of the activator. In this same binder, minor diffraction peaks correspond to the formation of sodium carbonate Na_2CO_3

(Ref. code: 96-152-7607), indicating that excess sodium hydroxide reacts with atmospheric CO_2 , leading to carbonation of the material.

The microstructure of the developed materials is next analyzed. Fig. 6 displays first SEM micrographs of the fracture surfaces of the 100CDW, 80CDW, and 60CDW materials with an activator modulus $M_s=1.0$ at 28 days of curing. It is observed that the microstructure of the control binder (100 CDW) is denser, composed of unreacted and heterogeneous precursor particles surrounded by gel. The inclusion of 20 wt% RHA leads to heightened porosity, a trend that diminishes slightly upon the inclusion of 40 wt% RHA, as will be further investigated in the examination of bulk density and total porosity data later in the study. In these 80CDW and 60CDW binders, fewer unreacted CDW

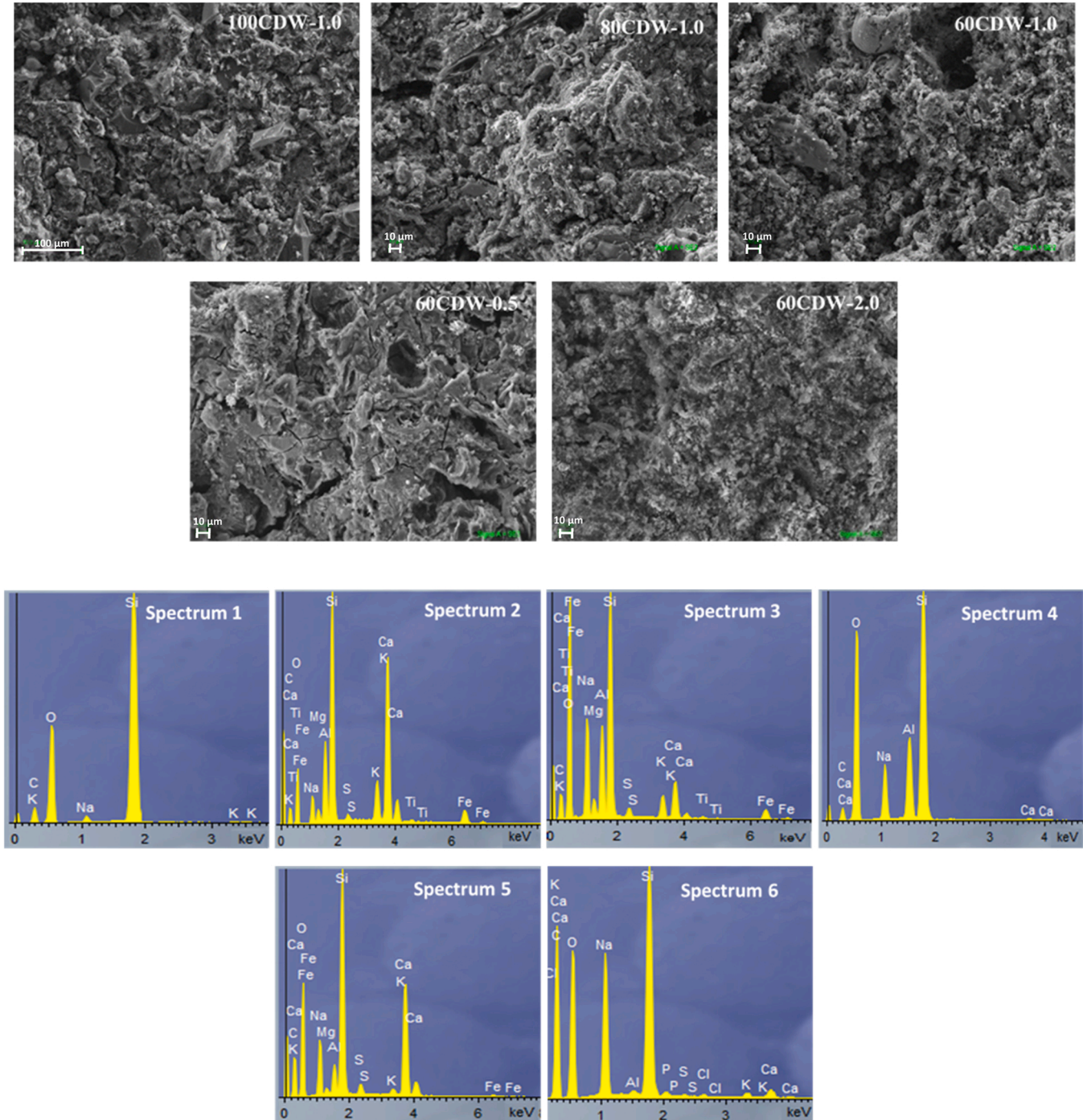


Fig. 6. SEM-EDS of alkali-activated materials of 100CDW-1.0, 80CDW-1.0, 60CDW 1.0, 60CDW-0.5 and 60CDW-2.0 at 28 days of curing.

and RHA particles were identified, with a higher amount of geopolymeric gel forming. Therefore, a priori, the higher amount of unreacted particles and the lower presence of gel in the 100CDW material may lead to structural defects and result in lower mechanical strength [73]. In particular, EDS analysis of the 100CDW binders shows the presence of embedded quartz particles in the matrix (Spectrum 1) and a mixture of C-(A)-S-H gels (Spectrum 2) and (N,C)-A-S-H type gels (Spectrum 3). The addition of increasing amounts of RHA leads to the formation of a larger amount of N-A-S-H geopolymeric gel (Spectrum 4), along with the formation of C-(A)-S-H gel (Spectrum 5). Furthermore, the employed activator modulus, Ms, also influences by altering the morphology of the binders. Thus, Fig. 6 summarizes the characteristic microstructure observed in the 60CDW-0.5 and 60CDW-2.0 materials at 28 days of curing. In the 60CDW-0.5 samples, a microstructure with a significant number of defects like pores and cracks in the matrix is observed. These defects are likely to originate during the manufacturing process due to the retention of air bubbles or shrinkage during the drying phase. Additionally, a higher number of unreacted or partially reacted particles are embedded in the matrix gel. Regarding the gel, a larger presence of (C,N)-(A)-S-H structure is observed, with a very limited quantity of sodium-rich N-A-S-H geopolymeric gel (Spectrum 6). The use of an activator modulus Ms = 1.0 and 2.0 results in the formation of a sodium aluminosilicate gel with a spongy and globular morphology, indicating a higher degree of geopolymerization in the samples. The use of a greater amount of alkali silicate enhances the effectiveness of dissolution and polycondensation reactions, leading to a larger quantity of geopolymeric gel capable of forming a more homogeneous structure, thus anticipating greater mechanical strength in these materials.

Fig. 7a illustrates the FTIR spectra of the precursors and the specimens manufactured with different percentages of RHA and an activator modulus Ms = 1.0, as well as the 60CDW materials with varying activator modulus Ms values (Fig. 7b), aiming to establish a comparison among them and discern the nature of the bonds formed during the alkali activation reaction. The assignment of the key spectral bands is summarized in Table 3. The FTIR analysis confirms the presence of the main “fingerprints” of the aluminosilicate matrix. It is observed that the band centered at 994 cm^{-1} and 1063 cm^{-1} in the CDW and RHA precursors, respectively, shifts towards higher wavenumbers in the alkali-activated cements, reaching 985 cm^{-1} in 100CDW⁻¹, 1020 cm^{-1} in 80CDW⁻¹, and 1006 cm^{-1} in 60CDW⁻¹ cements. These bands are attributed to the formation of asymmetric Si-O-T (T = Si or Al) bonds, namely sialate (Si-O-Al) and siloxo (Si-O-Si) bonds [74], exhibiting vibrational stretching modes. This band is significant for quantifying the degree of polycondensation in alkali-activated cements. The observed shift relative to the raw materials is associated with the partial depolymerization of the original tetrahedral structure of silicate and/or aluminosilicate precursors, followed by the polymerization of a new amorphous aluminosilicate gel, N-A-S-H, and a (N,C)-(A)-S-H gel [75, 76]. The higher area of this band in the 60CDW-1 specimens could indicate enhanced precursor dissolution and an increased amorphous character, attributed to a larger quantity of geopolymeric gel formation and thus an anticipated enhancement in mechanical strength [75, 76]. Furthermore, upon analyzing the influence of the activator modulus (Fig. 7b, Ms = 0.5, 1.0 and 2.0), it is observed that this band exhibits greater width when increasing the module, revealing that this parameter impacts the degree of geopolymerization of the hydrated gels due to increased dissolution of the amorphous phases present in the precursors. The bands around $440\text{--}790\text{ cm}^{-1}$ detected in all alkali-activated specimens stem from bending in the Si-O plane and the Al-O bonds, as well as the bending vibrations of the Si-O-Si and O-Al-O bonds [75, 77, 78].

There are other characteristic bands in the spectra, but they are not associated with changes in the aluminosilicate structure. The bands centered at $3400\text{--}3250\text{ cm}^{-1}$ and 1650 cm^{-1} are assigned to stretching and bending vibrations of H-O-H bonds of water molecules physically absorbed on the gel's surface. These bands are especially relevant in the

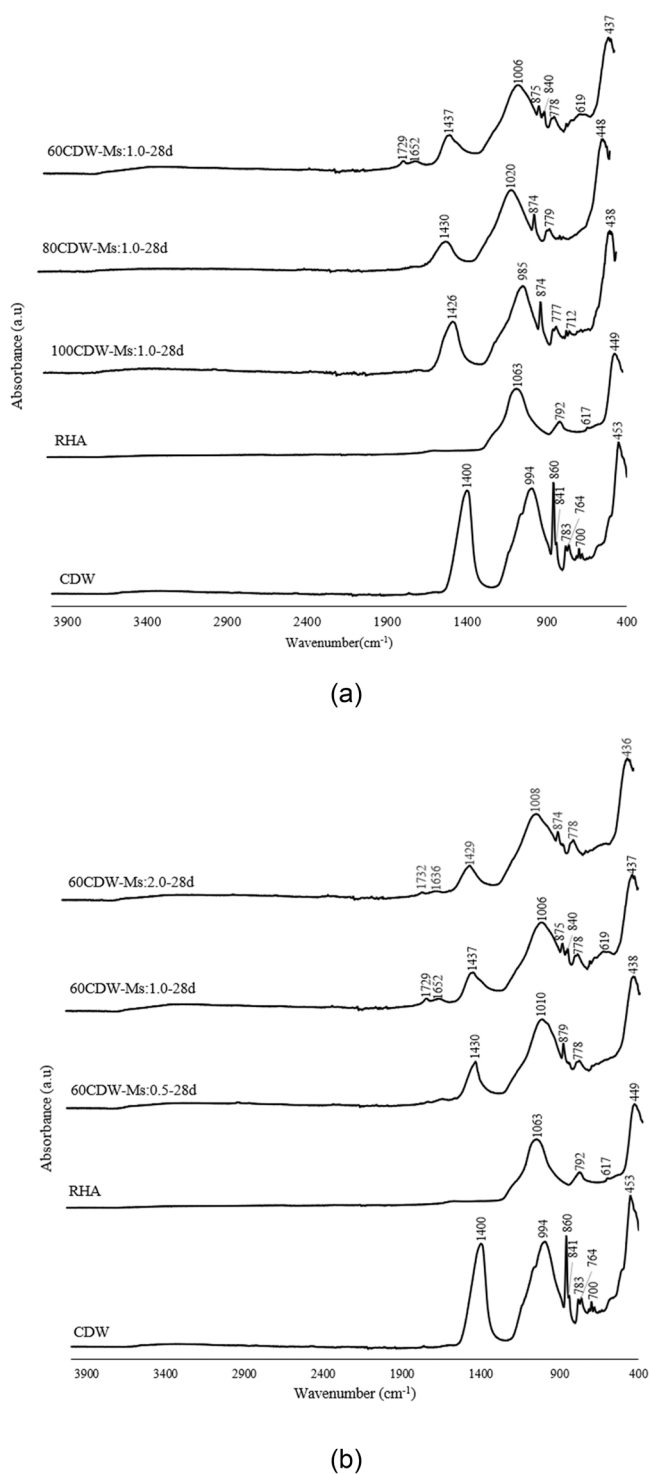


Fig. 7. a. FTIR of precursors and alkali-activated cements at 28 days of curing with Ms = 1.0. b. FTIR of precursors and 60CDW-40RHA at 28 days of curing with Ms = 0.5, 1.0 and 2.0.

case of the 60 CDW materials that were prepared with a larger amount of water. The bands centered at 1430 cm^{-1} and 874 cm^{-1} stem from the presence of calcite as well as atmospheric carbonation, attributed to asymmetric stretching and out-of-plane bending modes of CO₃. A difference in the intensity of the peaks for these latter bands it is observed in the different binders, diminishing in intensity as increasing amounts of RHA precursor are incorporated. Regarding the influence of Ms, there is higher intensity as Ms decreases due to an excess of sodium in the

Table 3
FTIR spectra band assignments.

Band number, cm^{-1}	Assignment
3400-3250 1650	H-O-H stretching and bending vibrations
1430, 875 1063-995	Atmospheric carbonation, CO_3 bending and stretching vibrations Si-O stretching vibrations and Si-O-Si or Al-O-Si asymmetric stretching vibrations
790-440	In plane Si-O bending and Al-O linkages as well as bending Si-O-Si and O-Si-O vibrations

activator, which can react with atmospheric CO_2 , leading to the formation of sodium carbonates in addition to calcium carbonates [75].

4.2. Porosity and physico-mechanical properties

After analyzing the microstructure and composition developed in the different materials, the ensuing sections present the physico-mechanical

properties and derived porosity resulting from these features. The data for bulk density after a 28-day curing period (Fig. 8a) reveal a slight reduction in bulk density upon introducing RHA into the mixture when utilizing a solution with $M_s = 0.5$. However, for $M_s = 1.0$ and 2.0 , the apparent density displays a decline with the addition of 20 wt% RHA and an increase with the inclusion of 40 wt% RHA. These outcomes can be primarily attributed to the lower real density of the RHA residue, measuring 2080 kg/m^3 , in comparison to the CDW residue's density of 2510 kg/m^3 . Furthermore, the water-to-binder ratio (w/b, Table 2) rises with the incremental incorporation of RHA due to the higher water absorption by the residue, essential for achieving the desired consistency and workability. However, there exists no excess liquid within the paste; any extra water is absorbed by the RHA residue, preventing a rise in porosity and pore count in the final products. In fact, upon integrating 40 wt% RHA with M_s values of 1.0 and 2.0, the bulk density experiences an increase, attributed to reduced porosity or enhanced matrix continuity in the alkali-activated material (Fig. 8b). This is the result from heightened reactivity, leading to increased formation of reaction

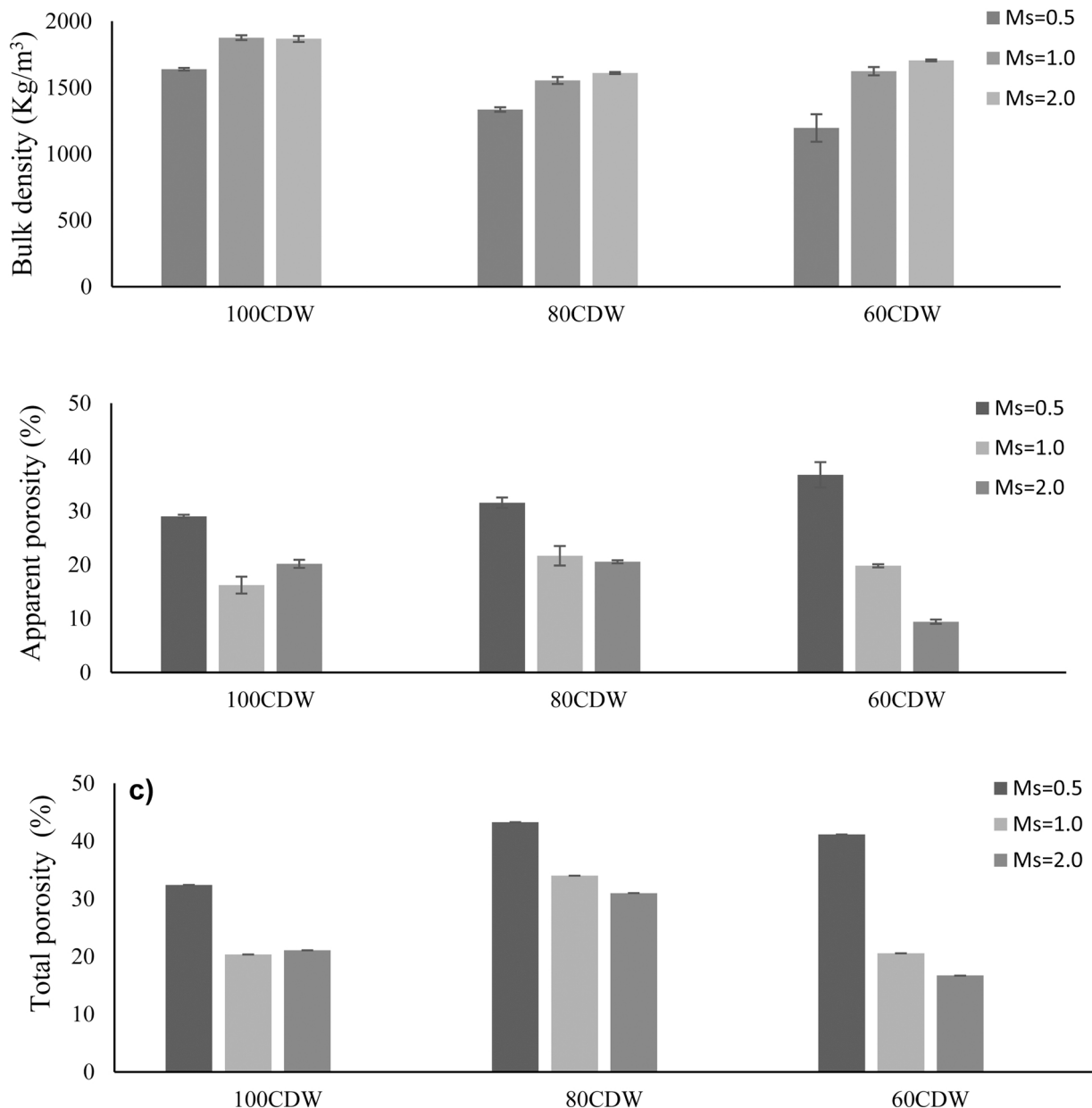


Fig. 8. Bulk density (a), apparent porosity (b) and total porosity (c) at 28 days of curing.

products that fill the pores.

Regarding the impact of Ms on bulk density (Fig. 8a), it becomes evident that the quantity of soluble silicate plays a significant role in the process of alkali activation. It is established that sodium silicate, on one hand, enhances the alkalinity of the liquid phase and expedites the dissolution of Si-Al constituents within the raw materials. This process involves a polycondensation reaction, resulting in the formation of N-(A)-S-H geopolymers, which contributes to an increase in specimen density. On the other hand, the heightened concentration of silicate ions within the liquid phase reacts with dissolved Ca^{2+} and Al^{3+} ions, leading to the creation of hydrated calcium aluminosilicate gels, C-(A)-S-H [72]. Consequently, the utilization of inappropriate quantities of alkaline solution and precursor (i.e. materials prepared with $\text{Ms}=0.5$) leads to less efficient dissolution of the precursors, giving rise to a more heterogeneous structure characterized by a greater number of pores within the matrix [79] as has been described from SEM observations in Fig. 6.

In fact, the apparent porosity results (Fig. 8b) exhibit a trend contrasting with that of bulk density. The values of apparent porosity were determined based on the water absorption test results, provided as supplementary material (Fig. S1). For the control binders (100CDW), the apparent porosity varies between 29.0% for an activator module $\text{Ms}=0.5$, and 16.2% and 20.2% for $\text{Ms}=1.0$ and 2.0, respectively. The addition of RHA under a low activator module ($\text{Ms}=0.5$) leads to an increase in apparent porosity, reaching values of up to 36.7 vol% for the 60CDW specimens. The greater availability of water (Table 2) when using $\text{Ms}=0.5$ results in higher porosity (lower density). When $\text{Ms}=1.0$ and $\text{Ms}=2.0$ activator modules are employed, the incorporation of 20 wt% RHA yields a slight increase in apparent porosity compared to the control material with the same activator module. Conversely, the addition of 40 wt% RHA leads to a turning point, causing a decrease in apparent porosity, reaching minimum values of 9.4% for $\text{Ms}=2.0$. As has been analyzed from SEM observations (Fig. 6) and FTIR characterization (Fig. 7b), the incorporation of 40 wt% RHA (series 60CDW) enhances gel formation that fills the pores. This reduction in apparent porosity results in a decreased water absorption capacity in the samples and in a foreseeable increase in its mechanical properties.

The results for total porosity are presented in Fig. 8c and have been computed considering the real density data, as shown in the supplementary material (Fig. S2). Main trends described for apparent porosity are similar to those observed for total porosity. Thus, it is evident that the addition of 20 wt% RHA leads to a notable increase in total porosity compared to the control samples, 100CDW, regardless of the employed

Ms. Conversely, the incorporation of 40 wt% RHA demonstrates a turning point, resulting in a decrease in total porosity. This reduction in total porosity reaches values equal to or lower than those of the control sample when an activator module $\text{Ms}=1.0$ and 2.0 are used. Regarding the influence of the activator module, within the CDW+RHA blends, lower total porosity is observed as the activator module is increased. This effect becomes more pronounced when transitioning from an activator module of 0.5 to 1.0. Accordingly, total porosity reaches values of up to 43.3% for the 80CDW binders and 41.1% for the 60CDW specimens when employing an activator module $\text{Ms}=0.5$. For activator modules $\text{Ms}=1.0$ and $\text{Ms}=2.0$, the influence of this parameter is less pronounced, resulting in total porosity values ranging from 34.0% to 31.0% for the 80CDW specimens, and from 20.6% to 16.7% for the 60CDW cements, with Ms values of 1.0 and 2.0, respectively. From this last reduction in total porosity, an increase in mechanical properties is expected. Thus, in a general way, the range of these porosity values is similar to that described for alkali-activated materials based on CDW and suitable for the development of materials with structural applications.

The flexural strength of the binders as a function of the wt% of RHA and the activator modulus, Ms, for curing times of 7 and 28 days is shown in Fig. 9. It can be observed that at both curing times, the incorporation of increasing amounts of RHA results in an increase in flexural strength. For instance, flexural strength values of 6.0 and 9.6 MPa were recorded after 28 days of curing for the 100CDW control specimens with activator modulus $\text{Ms}=1.0$ and 2.0, respectively. These flexural strength values increased to 9.2 and 13.5 MPa for $\text{Ms}=1.0$ and 2.0, respectively, with the addition of 40 wt% of RHA, representing an approximately 50% and 40% increase in flexural strength. This trend in materials with $\text{Ms}=1.0$ and 2.0 indicates that the polycondensation process improves with the incorporation of increasing amounts of RHA, a source rich in amorphous SiO_2 , leading to better bonding between the unreacted particles and the matrix, as well as the formation of a denser 3D N-(A)-S-H geopolymeric gel [80]. Regarding the influence of the activator modulus Ms, it can be observed that for low values of $\text{Ms}=0.5$, low flexural strengths develop: below 2.5 MPa for all specimens with 28 days of curing. Increasing the activator modulus, $\text{Ms}=1.0$ and $\text{Ms}=2.0$, leads to an increase in soluble silica due to the use of a higher amount of sodium silicate, resulting in greater activation of precursors and leading to a larger amount of N-A-S-H aluminosilicate gel and C-A-S-H gel due to the calcium content present in the CDW precursor, both gels contributing to the higher mechanical strength of the cements [81,82].

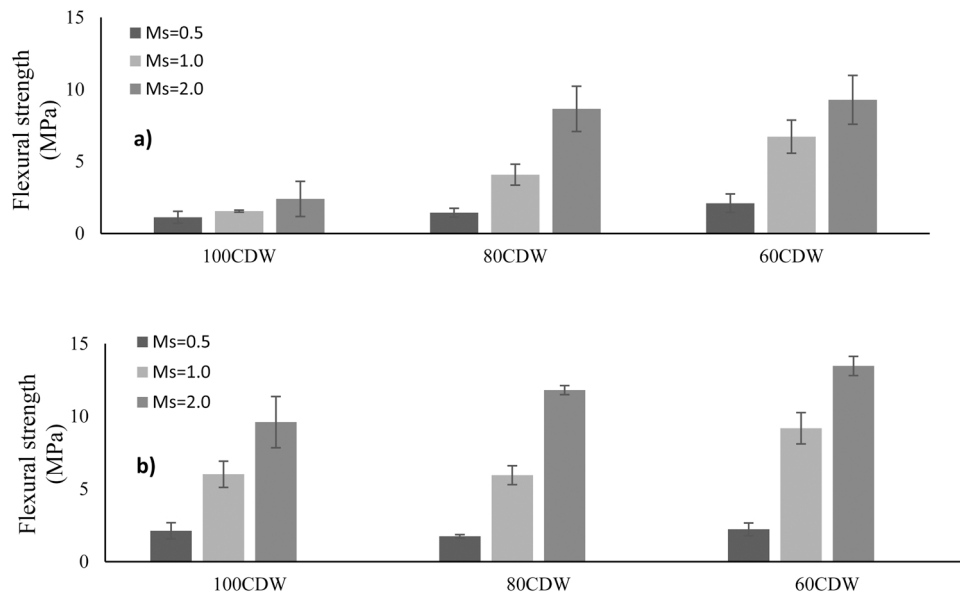


Fig. 9. Flexural Strength (MPa) at 7 days of curing (a) and 28 days of curing (b).

Fig. 10 displays the evolution of compressive strength at 7 and 28 days of curing. The trends observed with respect to RHA content and activator modulus are similar to those described for flexural strength values. The addition of RHA resulted in an increase in compressive strength for a 28-day curing period. Specifically, the incorporation of 20 wt% and 40 wt% of RHA allowed for achieving maximum compressive strength values of 41.0 MPa and 59.6 MPa for the 80CDW-2.0 and 60CDW-2.0 specimens, respectively. These values greatly exceed the resistances described by Tan et al. [47], for mixtures of CDW with Municipal solid waste fly ash and are in the order of magnitude of those described by Mahmoodi et al. [46], for mixtures of concrete waste (without ceramic fraction) with the addition of supplementary cementitious material such as fly ash, metakaolin or slag in percentages of 30–45 wt%.

Regarding the influence of curing time, samples with $M_s = 1.0$ and 2.0 show an increase in flexural strength with curing time (Fig. 9), with this increase being more notable in the control sample (100CDW). In other words, at early ages, a higher percentage of the final flexural strength develops in the samples incorporating RHA (i.e., the 80CDW and 60CDW samples with $M_s = 1.0$ and 2.0 develop around 70% of the 28-day strength during the first week of curing, compared to the 25% developed by the 100CDW samples). This could indicate a certain accelerating effect of the RHA residue in the alkali reaction, allowing geopolymerization or alkali activation to occur in a shorter time. However, establishing a clear trend of the effect on curing time based on the compressive strength test results (Fig. 10) proves more challenging. It is observed that the control binders 100CDW exhibit similar compressive strength values at both curing times, with maximum values for $M_s = 2.0$ being 17.3 and 18.5 MPa after 7 and 28 days of curing, respectively. This discrepancy between the temporal evolution of flexural and compressive behavior in CDW materials may highlight that the higher demands posed by mechanical bending stresses on ceramic materials translate, for this control composition, into reaction products that are more susceptible to this test during the initial stages of their formation than under compression conditions.

In general, for the mechanical properties of the developed materials, it can be assumed that an excessively low M_s ($M_s = 0.5$) negatively affects both flexural and compressive strength. A higher content of sodium silicate increases the alkalinity of the liquid phase and accelerates the dissolution of Si-Al components from the raw materials, leading to more extensive reaction and resulting in a greater quantity of monomers and

oligomers diffusing into the liquid phase and undergoing the geopolymerization reaction, forming N-A-S-H gels, thus increasing mechanical strength [83]. On the other hand, an increase in M_s raises the concentration of silicate ions in the liquid phase, which reacts with dissolved Ca^{2+} and Al^{3+} ions in the liquid phase to generate C-(A)-S-H gel products, resulting in greater density [69]. Therefore, the use of increasing amounts of sodium silicate ($M_s = 1.0$ and 2.0) and the RHA precursor contributes to a more rigid geopolymeric network structure, and additionally, it can ensure a better interface between the gel matrix and unreacted particles, thereby improving the mechanical performance of the cements [84].

Thermal conductivity is an important property when evaluating the insulation capacity of building products. In general, alkaline-activated materials have a significantly lower thermal conductivity than Portland cement, which has a conductivity of 1.5 W/mK [85]. Fig. 11 displays the thermal conductivity values obtained for the specimens produced after 28 days of curing. These values range between 0.24 and 0.72 W/mK, suggesting that, in addition to good mechanical properties, alkaline-activated materials with CDW and RHA also exhibit strong insulating capacity, thereby reducing associated energy costs. This property is closely linked to porosity and bulk density, such that lower porosity and higher bulk density values result in reduced insulation capacity and, consequently, higher thermal conductivity values. Specifically, the thermal conductivity values of 100CDW control cements range between 0.34 W/mK, when $M_s = 2.0$ is employed, and 0.47 W/mK for $M_s = 1.0$. The incorporation of 20 wt% RHA in the 80CDW mixture led to an increase in insulation capacity, as per total porosity data, yielding lower thermal conductivity values between 0.24–0.36 W/mK. However, the incorporation of 40 wt% RHA resulted in an increase in thermal conductivity, with values ranging from 0.32–0.62 W/mK. Regarding the influence of the activator modulus, it is observed that thermal conductivity increases as the M_s value is raised, in accordance with bulk density and total porosity data.

5. Conclusions

In this research, the synergistic effects of the addition of rice husk ash (RHA) (20–40 wt%) to construction and demolition waste (CDW) and the activator modulus ($M_s = 0.5$; 1.0 and 2.0) are investigated on the technological properties and microstructure of alkaline activation cements. The following conclusions can be drawn:

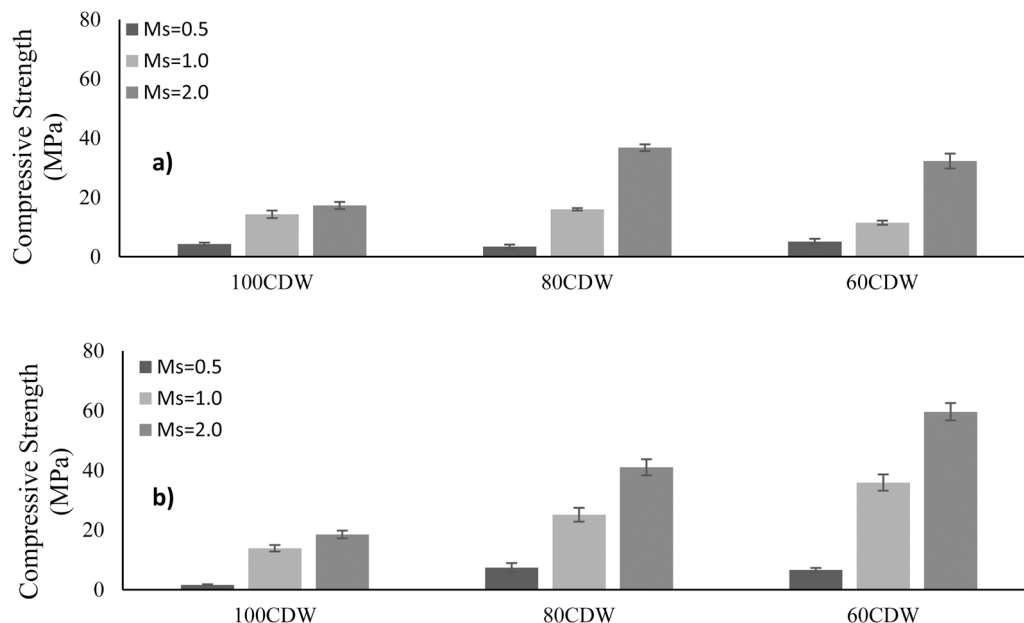


Fig. 10. Compressive Strength (MPa) at 7 days of curing (a) and 28 days of curing (b).

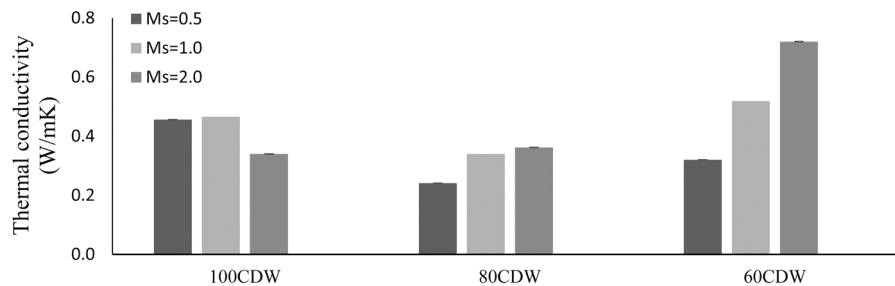


Fig. 11. Thermal conductivity of specimens at 28 days of curing.

1. The replacement of CDW by 20 wt% RHA produced a decrease in bulk density and an increase in total porosity and water absorption. However, the addition of 40 wt% of RHA promoted new gel formation and gave rise to alkaline activation cements with bulk density, water absorption and total porosity similar to the control binders. The activator modulus has a great influence on these properties, obtaining lower bulk density, and high total porosity and water absorption when a low activator modulus is used, $Ms = 0.5$. The use of a module $Ms = 1.0$ produced a significant increase in bulk density and a significant decrease in total porosity and water absorption. While the effect of increasing $Ms = 2.0$ was less significant.
2. The flexural and compressive strength results show that both the RHA/CDW ratio and the activator modulus are factors that must be considered simultaneously to achieve the desired mechanical properties. The maximum values are achieved for the highest incorporation of RHA, 40 wt% and the highest activator modulus $Ms = 2.0$, obtaining flexural and compression resistances after 28 days of curing of 13.5 and 59.6 MPa, respectively. Low activator modulus $Ms = 0.5$ results in relatively low mechanical strength. Therefore, both factors, the increase in RHA incorporation up to 40 wt% and the activator modulus up to $Ms = 2.0$, significantly increase the intensity of the alkaline activation reaction modifying the amount and structure of the formed gel.
3. The main reaction products in the control mixtures, 100CDW, are C-A-S-H type gels. The formation of this gel is attributed to the amount of calcium present in the precursor, available in the solution. As increasing amounts of RHA residue are incorporated, an influence is observed on the main structure of the gel, predominating the formation of N-A-S-H geopolymeric gel. As for the activator module, it has a great influence on the structure of the binders. Cements that use low modulus $Ms = 0.5$ have a more heterogeneous structure, with a greater amount of unreacted particles and greater porosity, with the (C,N)-(A)-S-H gel predominating. The use of modules $Ms = 1.0$ and 2.0 gives rise to a greater geopolymerization reaction, forming N-A-S-H gels, in addition to C-(A)-S-H gel due to greater dissolution of the reactive species.

The interesting mechanical properties obtained in this research indicate that construction and demolition waste, CDW, together with ashes from the combustion of rice husk biomass, RHA, are good candidates to be valued as precursors for alkaline activation cements. New future research is necessary to establish through Life Cycle Analysis the environmental improvement achieved in comparison with the environmental impacts associated with the production of Portland cement. In this sense, it is also of interest to evaluate the production of CDW/RHA binders from alternative alkaline activators based on wastes with high Si, Na or K content instead of commercial ones.

CRedit authorship contribution statement

Díaz García Almudena: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Data curation.
Pérez Villarejo Luis: Writing – review & editing, Validation,

Methodology, Funding acquisition, Formal analysis, Conceptualization.
Eliche-Quesada Dolores: Writing – review & editing, Writing – original draft, Validation, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.
Bueno Salvador: Writing – review & editing, Writing – original draft, Validation, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT / OpenAI in order to transcribe part of one's own written text (in the introduction part) into a more understandable language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.conbuildmat.2023.134823](https://doi.org/10.1016/j.conbuildmat.2023.134823).

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